

Classical Economic Thermodynamics: The Fundamental Laws of Motion of Value and Social Reproduction

Oriol Vallès Codina

April 25, 2026

We propose a two-law thermodynamic framework for classical political economy that resolves the question of whether economic systems admit genuine conservation laws. The first law: the Monetary Equivalent of Labour Time (MELT) provides an aggregate conservation identity $\Pi = S$ holding for any price system — total monetary profit equals total surplus value — proven in full generality by Shaikh (2016). This is a contingent empirical result, not a tautology. The second law: labour is an irreversible thermodynamic process. Abstract labour — the socially homogeneous quantum of human metabolic energy expenditure — is the economic expression of this metabolic irreversibility; the MELT converts it to monetary value, playing the role of Boltzmann’s constant at the interface between thermodynamic substrate and monetary form. We derive Hamiltonians for three classical economic models — Goodwin’s predator-prey class dynamics, Flaschel and Semmler’s multi-sector competitive dynamics, and Foley’s circuit of capital — and prove that all three satisfy $\dot{H} = 0$ under simple reproduction (constant capital stock), with explicit position-momentum-velocity-force interpretations in each case. For expanded reproduction, we derive the unified formula $\dot{\mathcal{H}} = s_c \varepsilon L / (1 + \varepsilon)$ across all three frameworks, where ε is the exploitation rate, L is living labour, and s_c is the capitalists’ savings rate. The circuit Hamiltonian $\mathcal{H} = K/\mu$ is total capital in labour-time units; $\dot{\mathcal{H}}$ is economic power in the thermodynamic sense — the rate at which the second law drives the first-law conservative manifold outward. The neoclassical conservation programme fails because its only invariant — the Hicksian expenditure function — exists in unobservable compensated demand space; the Sonnenschein-Mantel-Debreu theorem eliminates any aggregate invariant in observable space.

1 Introduction

The question of whether economic systems possess genuine conservation laws has a long history in both physics and economics. Mirowski’s *More Heat than Light* (1989) argued that the marginal revolution of the 1870s was a systematic transplantation of classical mechanics into economic theory: Walras modelled exchange as force balance, marginal utility as generalised force, and equilibrium as

energy minimum. The programme sought an economic analogue of energy conservation. Mirowski concluded it had failed.

This paper argues the failure was a failure of analogy, not of programme. The correct physics analogy for a capitalist economy is not classical mechanics but thermodynamics — and specifically, both laws of thermodynamics are required, not just the first. A mechanical system is closed, time-reversible, and self-sustaining: it requires no external input. A capitalist economy is none of these. It is open (labour is an external input), irreversible (production produces entropy), and requires continuous expenditure of living labour to reproduce itself. These are second-law properties. An economic theory that ignores the second law cannot provide a genuine conservation law, because it cannot identify the source of the conserved quantity.

The physics literature has long recognised economic systems as candidates for thermodynamic and Hamiltonian analysis. Georgescu-Roegen’s *The Entropy Law and the Economic Process* (1971) was the first systematic attempt to apply the second law to economics, but it detached entropy from labour value, treating thermodynamic limits as external constraints rather than as the constitutive basis of value itself. Foley and Smith’s Santa Fe lectures (2007) identified the competitive equilibrium as the economic analogue of the maximum-entropy state in statistical mechanics, and derived the MELT as the bridge between the thermodynamic substrate and the monetary form. The econophysics literature has developed statistical mechanics approaches to financial markets and income distributions but has not, to our knowledge, proposed a complete two-law thermodynamic framework grounded in classical political economy. The present paper fills this gap.

The present paper makes four contributions:

1. **Two conservation laws** for classical political economy: a first-law analogue (the MELT conservation identity, holding for any price system) and a second-law analogue (labour as irreversible thermodynamic work)
2. **Hamiltonian derivations** for three classical models (Goodwin, Flaschel-Semmler, Foley circuit) with explicit position-momentum-velocity-force mappings
3. **A unified formula** for $\dot{\mathcal{H}}$ in expanded reproduction across all three frameworks
4. **A precise diagnosis** of why the neoclassical conservation programme fails (confinement to unobservable Hicksian space)

The paper is organised as follows. Section 2 states the failure of the neoclassical conservation programme. Section 3 presents the classical conservation principle. Section 4 develops the first law (MELT identity). Section 5 derives the three Hamiltonians and the unified $\dot{\mathcal{H}}$ formula. Section 6 develops the second law. Section 7 treats expanded reproduction and the secular entropy maximum.

2 The Neoclassical Conservation Programme and Its Failure

Walras modelled exchange explicitly as force balance: marginal utility as generalised force, quantities as displacements, and equilibrium as energy minimum. The differential relation $dU = \sum_i p_i dx_i$ implied a potential function with demand as its gradient. Slutsky symmetry guaranteed compensated substitution effects form a conservative vector field. The programme sought to build a first conservation law on this foundation.

Before locating the failure, one structural feature deserves note. The budget constraint $p \cdot x = m$ — Walras’s Law $p \cdot z(p) = 0$ for excess demand — is a *zeroth-law* condition accepted by both traditions: it grounds prices as state variables and establishes the transitivity of exchange. The dispute is not about the zeroth law but about what first law can be built upon it.

The neoclassical programme attempts to build the first law on utility. The **Hicksian expenditure function**

$$e(p, \bar{u}) = \min_x \{p \cdot x : u(x) \geq \bar{u}\}$$

is the potential of a conservative vector field: $x_i^H = \partial e / \partial p_i$, so the Hicksian substitution matrix $\partial x_i^H / \partial p_j$ is symmetric negative semidefinite. But Hicksian demand $x^H(p, \bar{u})$ holds utility $\bar{u}$ fixed — it is *unobservable*, defined only relative to a reference utility level that cannot be measured. Observable **Walrasian (Marshallian) demand** $x^M(p, m)$ holds nominal income m fixed; the Slutsky decomposition

$$\frac{\partial x_i^M}{\partial p_j} = \frac{\partial x_i^H}{\partial p_j} - x_j^M \frac{\partial x_i^M}{\partial m}$$

shows that income effects break symmetry in the Marshallian matrix. The conservative field exists only in unobservable, distributionally-frozen Hicksian space. Afriat’s Theorem (1967) gives the conservation an empirical handle through GARP — but GARP tests integrability, not the ontological claim that utility is energy. The neoclassical “conservation principle” reduces to: budget constraint plus preference consistency yield a conservative expenditure field in unobservable space. This is not energy conservation; it is a tautological restatement of budget feasibility in potential-function language.

At the aggregate level, the Sonnenschein-Mantel-Debreu theorem establishes that any continuous function satisfying Walras’s Law is admissible as aggregate excess demand: no regularity condition from individual optimisation survives aggregation. No macro-level conservation survives. The Cambridge capital controversies (Cohen and Harcourt 2003) administered a structural blow: the neoclassical aggregate production function $Y = F(K, L)$ requires capital to be measurable in physical units independent of distribution — but the value of heterogeneous capital goods depends on

the rate of profit, which the marginal product is supposed to explain. This circularity (the Wicksell effect) generates reswitching and capital-reversing, acknowledged as logical necessities. There is no consistent aggregate capital measure; the neoclassical programme cannot specify the invariant it would need to conserve.

A further observation: the neoclassical programme never engaged with the *second law* at all. Utility theory has no concept of irreversibility — a consumer could, in principle, reverse every decision without loss. Production is a reversible transformation. There is no entropy, no metabolic cost, no thermodynamic limit. The second law is simply absent. An economics without irreversibility has no theory of why production requires labour, or why ecological limits exist.

3 The Classical Conservation Principle

3.1 Smith and the Labour Command

Adam Smith’s *Wealth of Nations* (1776) contains, despite its inconsistencies, the germ of the conservation principle. In the labour-embodied reading of Smith’s dual account, the price at which a commodity exchanges is regulated in the long run by the labour required for its production. Exchange redistributes the product of labour; it does not create value. The “real price” of everything is the toil and trouble of its production.

3.2 Ricardo and the Invariable Measure

David Ricardo sharpened this into a systematic claim: relative exchange values are regulated by relative labour times, and changes in the wage-profit distribution leave them approximately unchanged — his “93 per cent” approximation. But Ricardo was honest enough to see the problem: a rise in wages alters relative prices differentially, depending on the capital intensity of each sector, so that no produced commodity can serve as an invariable measure of value immune to distributional changes. Ricardo died in 1823 believing this problem insoluble.

The analytical content of the “93 per cent” is now understood precisely through the price-of-production formula: at $r = 0$, prices are exactly proportional to labour values; at $r = R$, prices are proportional to the standard commodity. For empirically observed profit rates (well below R), the deviation from labour-value proportionality is bounded and small. The “93 per cent” is not a casual approximation but a structural result: the Wicksell effects are real but compressed because the capital intensity $\Omega_k = C_k/V_k$ does not vary wildly across sectors in practice.

3.3 Marx and the Conservation Theorem

Marx resolved the confusion between concrete and abstract labour that had trapped Ricardo. *Concrete labour* is the specific activity performed — weaving, smelting, programming. *Abstract labour* is what all concrete labours share: the expenditure of human labour-power, stripped of its specific form, validated in exchange as socially necessary. Abstract labour constitutes the substance of value.

The conservation principle follows with precision. In exchange, commodities are equated as values: both contain the same quantity of abstract labour. Exchange creates no value; it only realises it. Shaikh (1977) states this formally: “no value is created in the circulation process” — citing Marx’s own formulation in *Capital* (Vol. II): whatever prices prevail, the plus on one side is a minus on the other. The total value distributed is fixed by what was created in production. This is a contingent empirical claim, not a tautology.

The **exploitation rate**

$$\varepsilon = \frac{\mu}{w} - 1 \quad \Longleftrightarrow \quad \mu = w(1 + \varepsilon), \quad \boxed{\varepsilon = \frac{S}{V} = \frac{\text{surplus labour time}}{\text{necessary labour time}}}$$

where μ is the MELT and w is the money wage, measures the ratio of surplus to necessary labour time. The sectoral profit rate $\rho_k = \varepsilon_k / (1 + \Omega_k)$ shows that differential capital intensities Ω_k generate differential profit rates even at uniform exploitation — the classical source of price-value deviations and the foundation of Sraffa’s prices of production.

4 First Law: Value is Conserved in Exchange

4.1 Labour Values and the Price-Value Map

Let A be the $n \times n$ input-output matrix, ℓ the labour coefficients vector, and x the gross output vector. Labour values v are the total (direct + indirect) labour requirements per unit output:

$$v = vA + \ell \implies v = \ell(I - A)^{-1}$$

The **MELT** is defined as the ratio of the money value of the net product to total living labour:

$$\mu \equiv \frac{p^\top (I - A)x}{\ell^\top x} = \frac{Y}{L}$$

where $Y = p^\top(I - A)x$ is the money value of net output and $L = \ell^\top x$ is total living labour in hours. The MELT converts labour-time to money: it is the “price” of an hour of abstract labour in monetary units.

The price-value map decomposes prices of production p^* into three components (Shaikh 2016, Ch. 6):

$$p^*(r) = \underbrace{v\mu}_{\text{labour values}} + r \underbrace{v\left(H - \frac{1}{R}I\right)\mu}_{\text{Marxian term}} + r \underbrace{[p^*(r) - v]H\mu}_{\text{Wicksell-Sraffa term}}$$

The *Marxian term* is proportional to the deviation of each sector’s capital intensity from the standard commodity; the *Wicksell-Sraffa term* is the feedback from price-value deviations onto the capital valuation. Both vanish at $r = 0$, where $p^* = v\mu$. Their empirical magnitude determines whether prices are “close” to values — and Shaikh (2016), Işıkara and Mokre (2021), and Tsoulfidis and Maniatis (2008) find that cross-sectoral price-value deviations are small (mean absolute deviation $\sim 10\text{--}15\%$), consistent with the structural boundedness of the Marxian and Wicksell-Sraffa terms.

Prices of production follow explicitly:

$$p^* = (1 - r/R) v (I - Rr/R H)^{-1}$$

where $R = 1/\lambda_1 - 1$ is the maximum rate of profit (λ_1 the dominant eigenvalue of A) and $H = K(I - A)^{-1}$. At $r = 0$: $p^* = v$ (prices proportional to labour values). At $r = R$: p^* proportional to the standard commodity — the leading eigenvector of A and Ricardo’s invariable measure of value.

4.2 The Universal Conservation Identity

The First Law of Classical Economic Thermodynamics (Shaikh 2016): For *any* price system p — market prices, prices of production, monopoly prices, or any other:

$$\boxed{\mu L = p^\top(I - A)x \quad \Longleftrightarrow \quad \Pi = S}$$

Total monetary profit Π equals total surplus value S for any price system. This follows directly from the definition of MELT: multiplying $\mu = Y/L$ through gives $\mu L = Y = p^\top(I - A)x$. The decomposition $Y = W + \Pi$ (wages plus profit) and $L = V_\ell + S_\ell$ (necessary plus surplus labour) gives $\Pi = \mu S_\ell = S$. The proof requires no assumption about which price system prevails. It holds at market prices, at prices of production, and at any other price system — including monopoly

prices. Whatever redistribution occurs in exchange (the plus on one side is a minus on the other), the aggregate conserves. This is the first law.

The MELT μ plays the role of Boltzmann's constant: it converts between the thermodynamic substrate (hours of abstract labour) and the observable monetary form. Just as k_B converts between microscopic kinetic energy and macroscopic temperature, μ converts between labour-time and money-value. The conservation identity is the economic expression of this conversion: the money value of the net product is the monetary form of the total abstract labour irreversibly expended in its production.

5 Conservative Hamiltonian Dynamics: Three Models

5.1 The Hamiltonian: Definition

In classical mechanics, $H(q, p)$ is a scalar function of generalised coordinates q (positions) and momenta p . Hamilton's equations $\dot{q}_i = \partial H / \partial p_i$ and $\dot{p}_i = -\partial H / \partial q_i$ generate the dynamics. If H has no explicit time dependence, $\dot{H} = \partial H / \partial t = 0$: energy is conserved. When external forces inject or remove energy, $\dot{H} \neq 0$.

Three conditions generate $\dot{H} = 0$ in the economic context:

1. **Fixed technique:** A, ℓ, μ constant $\Rightarrow \partial H / \partial t = 0$
2. **Constant scale:** $\dot{K} = 0$ — no source term from net investment
3. **Frictionless competition:** $\Delta_p = \delta_p I$ — uniform adjustment speeds preserve the symplectic structure

5.2 Goodwin's Growth Cycle

Goodwin's predator-prey model couples employment rate λ and wage share ω :

$$\dot{\lambda} = \lambda(a - b\omega), \quad \dot{\omega} = \omega(-c + d\lambda)$$

In log-coordinates $q = \ln \lambda$, $p = \ln \omega$, so $\dot{q} = a - be^p$ and $\dot{p} = -c + de^q$. These are Hamilton's equations $\dot{q} = \partial H / \partial p$, $\dot{p} = -\partial H / \partial q$ for:

$$H(q, p) = ap - be^p + cq - de^q, \quad \dot{H} = 0$$

One verifies: $\partial H/\partial p = a - be^p = \dot{q}$ and $-\partial H/\partial q = -(c - de^q) = -c + de^q = \dot{p}$. The equilibrium is $(\lambda^*, \omega^*) = (c/d, a/b)$; the exploitation rate at equilibrium is $\varepsilon^* = (1 - \omega^*)/\omega^*$.

Position-momentum-velocity-force mapping:

Term	Mechanical	Goodwin
$q = \ln \lambda$	Position	Employment: capital's deployment of living labour
$p = \ln \omega$	Momentum	Wage share: workers' accumulated distributional power
$\partial H/\partial p = a - b\omega$	Velocity	Net accumulation rate (wage-dampened employment growth)
$-\partial H/\partial q = -c + d\lambda$	Force	Phillips curve: employment pressure on wages

$H = (ap - be^p) + (cq - de^q)$: wage-momentum energy plus employment-position energy. The conserved H is the **intensity of class struggle** — the amplitude of the distributional orbit in (λ, ω) space, conserved because neither capital nor labour wins decisively. The three simple-reproduction conditions ensure $\dot{H} = 0$: fixed technique gives constant a, b, c, d ; constant scale prevents $\dot{K}$ entering $\dot{q}$; frictionless competition preserves the Lotka-Volterra structure.

5.3 Flaschel-Semmler Multi-Sector Dynamics

Flaschel and Semmler (1987) extend the Hamiltonian structure to an n -sector economy. Let $M = B - RA$ where $B = I - A$ and R is the maximum profit factor. With $y = \ln p$, $z = \ln x$ and uniform adjustment coefficients δ_p, δ_x :

$$\dot{y} = -\delta_p M e^z, \quad \dot{z} = \delta_x M^\top e^y$$

These are Hamilton's equations for:

$$H(z, y) = \delta_x \mathbf{1}^\top M^\top e^y + \delta_p \mathbf{1}^\top M e^z, \quad \dot{H} = 0$$

The equilibrium is the Sraffian long-period position: prices at p^* , quantities at standard commodity proportions.

Position-momentum-velocity-force mapping:

Term	Mechanical	Flaschel-Semmler
$z_k = \ln x_k$	Position	Sector k output: capital's deployment across industries
$y_k = \ln p_k$	Momentum	Sector k price: accumulated profitability signal
$\partial H / \partial y_k$	Velocity	Classical law of excess profitability (price above cost $\rightarrow$ output rises)
$-\partial H / \partial z_k$	Force	Walrasian law of excess demand (output below equilibrium $\rightarrow$ price rises)

$H = \delta_x \mathbf{1}^\top M^\top e^y + \delta_p \mathbf{1}^\top M e^z$: kinetic value (price-momentum, profit differentials driving capital flows) plus potential value (quantity-position, output imbalances driving prices). The conserved H is the **intensity of inter-sectoral competitive tension**.

Key contrast with Goodwin: Goodwin operates at the **class level** (labour vs. capital in aggregate); FS operates at the **sectoral level** (capital vs. capital across industries). Goodwin H is class struggle intensity; FS H is competitive tension. Both are first-law redistribution dynamics — closed orbits with no net creation or destruction of value. When the uniform adjustment condition $\Delta_p = \delta_p I$ fails — due to markup pricing, non-uniform capital mobility, or financial barriers — $\dot{H} \neq 0$ locally but $\oint \dot{H} dt = 0$ over a full cycle: the frictions break exact conservation while preserving it on average.

5.4 Foley's Circuit of Capital

The circuit consists of three stocks — productive capital $N(t)$, commercial capital $X(t)$, financial capital $F(t)$ — with evolution:

$$\dot{N} = C - P, \quad \dot{X} = P - S', \quad \dot{F} = S - (1 - p)S'' - C$$

Summing directly: $\dot{K} = pS''$. Via the MELT ($S'' = \mu S_\ell$), define the **circuit Hamiltonian** $\mathcal{H} \equiv K/\mu$:

$$\boxed{\dot{\mathcal{H}} = \frac{\dot{K}}{\mu} = p S_\ell = \frac{p\varepsilon}{1 + \varepsilon} L}$$

$\mathcal{H} = K/\mu$ is **total capital in abstract-labour-time units** — the economy's capacity to command living labour, measured in the thermodynamic substrate. $\dot{\mathcal{H}}$ is economic power (labour-hours per period). In simple reproduction ($p = 0$): $\dot{\mathcal{H}} = 0$.

The circuit Hamiltonian maps onto the mechanical analogy differently from Goodwin and FS. Rather than a two-dimensional phase-space Hamiltonian with canonical (q, p) coordinates, $\mathcal{H} = K/\mu$ is the conserved scalar of the circuit's aggregate dynamics:

Term	Mechanical	Foley circuit
$\mathcal{H} = K/\mu$	Total energy	Capital stock in labour-time: economy's capacity to command labour
$\dot{\mathcal{H}} = 0$	Closed system	Simple reproduction: circuit closes at constant scale
$\dot{\mathcal{H}} = pS_\ell$	External power input	Expanded reproduction: surplus labour reinvested
$\dot{\mathcal{H}} < 0$	Energy dissipation	Crisis: capital devaluation, anticipated values unrealised
Turnover rate $1/(T_P + T_R + T_F)$	Velocity	How fast value completes the full circuit
Profit rate $r = q/(T_P + T_R + T_F)$	Driving force	Markup that makes the circuit self-sustaining

The distinction from Goodwin and FS: those are *distributional* Hamiltonians (measuring phase-space amplitude of class or competitive orbits); $\mathcal{H} = K/\mu$ is the *stock* Hamiltonian (measuring the total value of the capital stock in thermodynamic units). In simple reproduction, all three are conserved. In expanded reproduction, all three drift upward at the same rate $s_c \varepsilon L / (1 + \varepsilon)$ — the second law asserting itself through living labour.

5.5 Frictions and Local Non-Conservation

The three conditions for $\dot{H} = 0$ define an ideal. Real economies operate with frictions that break condition 3 (frictionless competition: $\Delta_p = \delta_p I$) while preserving conservation on average. The canonical case is Foley's (1987) liquidity-profit rate cycle: firms hold fraction λ of wealth in liquid form; the interaction between profit rate π and liquidity preference generates:

$$\dot{\pi} = \alpha \pi (\bar{u} - u(\pi, \lambda)), \quad \dot{\lambda} = \beta (\pi - r_\lambda)$$

where r_λ is the return on liquid assets. At equilibrium, the Jacobian has purely imaginary eigenvalues — closed orbits analogous to Goodwin. When credit expansion reduces r_λ , a **Hopf bifurcation** occurs: a limit cycle emerges in the (π, λ) plane. Locally $\dot{H} \neq 0$, but $\oint \dot{H} dt = 0$ over the full cycle. The MELT identity is temporarily violated (credit money exceeds μL) and restored by the financial contraction.

Taking log-coordinates $q = \ln \lambda$, $p = \ln \pi$: *position* q measures where capital stands on the liquid-to-productive spectrum; *momentum* p is the profit rate — accumulated competitive pressure driving capital out of liquid positions. The Hopf bifurcation is the breakdown of the conservative orbit through a monetary friction — the simplest model of how financial dynamics break exact conservation while preserving it on average. Further frictions (markup pricing, non-uniform capital mobility, wage rigidities) operate by the same mechanism, each breaking condition 3 while leaving the long-run attractor (p^* , standard commodity proportions) intact.

5.6 The Unified $\dot{\mathcal{H}}$ Formula

Model	$\dot{H}$ — simple reproduction	$\dot{H}$ — expanded reproduction
Goodwin	0	$s_c \varepsilon L / (1 + \varepsilon)$
Flaschel-Semmler	0	$s_c \varepsilon L / (1 + \varepsilon)$
Foley circuit	0	$p \varepsilon L / (1 + \varepsilon)$

with $p = s_c$ by the Cambridge equation ($g = s_c r$). The time derivative of the Hamiltonian is uniquely determined by the surplus labour time and the reinvestment rate, regardless of which representation is used. This is a strong consistency result: the three models — operating at the class level (Goodwin), sectoral level (FS), and financial circuit level (Foley) — all have the same thermodynamic signature in expanded reproduction. The MELT is the invariant that bridges them. The exploitation rate ε is the **thermodynamic intensity** of expanded reproduction; s_c is the fraction of the thermodynamic drive converted into capital expansion rather than capitalist consumption.

6 Second Law: Labour as Irreversible Thermodynamic Process

Every act of production involves three irreversible thermodynamic processes running simultaneously.

Metabolic expenditure. The worker burns calories, depletes ATP, fatigues muscles and nerves. Chemical potential energy (food) is converted to mechanical and cognitive work and then to heat. This is a one-way process: the worker cannot un-expend the metabolic energy. Rest and food restore labour-power, but at thermodynamic cost — more entropy produced in the reproduction of the worker than in the work itself.

Material transformation. Ore becomes steel, cotton becomes cloth, silicon becomes a chip. These transformations are irreversible: the second law forbids spontaneous reassembly. The entropy of the material system increases with every productive transformation.

Temporal expenditure. The working day consumed is gone. Labour-time is the most irreversible resource in the economy: it cannot be stored, recovered, or reversed. The MELT μ is a *flow* concept — value per unit time — precisely because time is the dimension along which labour is irreversibly expended.

The commodity produced is a local pocket of order — low internal entropy. But this local order is paid for by greater disorder elsewhere: in the worker’s metabolic system, in depleted raw materials, in waste heat. This is the second law exactly: local entropy decrease (the ordered commodity) accompanied by global entropy increase (metabolic cost + material degradation + thermal dissipation). The commodity’s value is the social accounting of the total irreversible thermodynamic cost of its production.

Abstract labour as thermodynamic quantity. The distinction between concrete and abstract labour is, read thermodynamically, the distinction between the specific thermodynamic process and the abstract quantity of entropy production it involves. *Concrete labour* is the specific irreversible process — the particular chemical reactions, mechanical motions, and cognitive operations of weaving, smelting, programming. *Abstract labour* is the homogeneous quantum of human metabolic energy expenditure stripped of its specific form. What all concrete labours share is that they consume human bioenergy irreversibly at a rate measurable in time. The social reduction of concrete to abstract labour is the social recognition that all production involves the same kind of thermodynamic process. The MELT μ is the conversion factor between this thermodynamic quantity (hours of irreversibly expended human energy) and its monetary expression — like Boltzmann’s constant k_B , which converts between temperature (macroscopic, observable) and mean molecular kinetic energy (microscopic, thermodynamic substrate). The conservation identity $\mu L = p^\top (I - A)x$ acquires its full interpretation: the monetary value of the net product equals the social accounting of the total irreversible metabolic expenditure required to produce it.

The relation to the Hamiltonian framework. The second law is what makes the first law possible. Simple reproduction conserves value in exchange ($\dot{H} = 0$) precisely because living labour is continuously expended — replacing the consumed means of production exactly each period. The expenditure is irreversible (second law) but its monetary equivalent is exactly recovered through exchange (first law). Expanded reproduction ($\dot{H} > 0$) requires living labour to produce more than is needed for replacement: the second law drives the first-law manifold outward. Without irreversible labour expenditure, there is no value substrate for conservation to operate on.

7 Expanded Reproduction and Non-Conservative Dynamics

When $p > 0$ (capitalists reinvest a fraction of surplus value), the three conditions for conservation are systematically violated. Living labour injects new value each period; the Hamiltonians drift. Shaikh’s macro framework (Shaikh 2016) organises this non-conservation into three nested dynamic layers, each corresponding to a distinct pattern of $\dot{\mathcal{H}}$.

7.1 The Goodwin Layer: Turbulent Redistribution (~5 years)

At the shortest horizon, the Goodwin dynamics provide the structural skeleton. In Shaikh’s formulation, competition is “turbulent” — active profit-seeking through cost reduction and price-cutting — so $\dot{H}^{\text{Goodwin}} \approx 0$ with small dissipative fluctuations. The Goodwin orbit is approximately conservative; the turbulence is its envelope. The unified formula gives $\dot{\mathcal{H}} = s_c \varepsilon L / (1 + \varepsilon)$ on average, rising during expansion and falling during contraction.

7.2 The Juglar Layer: Investment Cycles (~9 years)

At the medium horizon, the Juglar fixed-capital cycle generates genuine non-conservation. Let g_K be the capital growth rate and r the realized profit rate:

$$\dot{g}_K = \theta (r - r^*), \quad \dot{r} = -\sigma g_K + \text{const}$$

where θ governs investment responsiveness to above-normal profitability and σ captures how expanding capacity compresses profit rates. The system oscillates in (r, g_K) space. During expansion: $\dot{\mathcal{H}} > 0$ — new investment mobilises new living labour, the conservative manifold expands. During contraction: $\dot{\mathcal{H}} < 0$ — capital is written off, anticipated values go unrealised. Unlike the Foley financial cycle (conservative on average), the Juglar involves real capital creation and destruction — genuine non-conservation at medium frequency.

7.3 The Long Wave: Secular Entropy Maximum (~50 years)

At the longest horizon, Marx-biased technical change raises the capital-labour ratio $\kappa = C/\ell$ (capital per unit of living labour):

$$\dot{\kappa} = (g_K - g_L - \delta) \kappa$$

The equilibrium profit rate is:

$$r^* = \frac{\varepsilon}{1 + \varepsilon} \cdot \frac{1}{\kappa}, \quad \frac{dr^*}{d\kappa} = -\frac{\varepsilon}{(1 + \varepsilon)\kappa^2} < 0$$

Rising κ displaces living labour from production — dead labour (machines) substitutes for living labour — and r^* falls secularly. The **Marxian source** of the secular decline: technical change simultaneously raises productivity and displaces the source of surplus value.

The **Ricardian source** operates from outside: exhaustion of natural entropy sinks (atmospheric CO₂ absorption capacity, soil fertility, aquifer recharge) raises ground rent — the price of accessing external entropy sinks — compressing profitability from outside the production sphere. Both sources drive r^* toward zero:

$$\dot{\mathcal{H}}_{\text{secular}} = s_c \frac{\varepsilon}{1 + \varepsilon} L \rightarrow 0 \quad \text{as} \quad r^* \rightarrow 0$$

The **secular entropy maximum** is the asymptotic limit of capitalist accumulation: the point at which the surplus available for reinvestment vanishes, approached as the system simultaneously exhausts its source of value (living labour, rising κ) and fills its entropy sink (nature, rising ground rent). In thermodynamic terms: the economy's capacity to extract low-entropy inputs and expel high-entropy outputs — the precondition for the irreversible labour process — is itself bounded. When both boundaries are approached simultaneously, $\dot{\mathcal{H}} \rightarrow 0$ and the conservative manifold can no longer expand.

The three nested layers together constitute Shaikh's hierarchical Hamiltonian structure:

Layer	Timescale	$\dot{\mathcal{H}}$ character	Mechanism
Goodwin	~5 years	≈ 0 (turbulent)	Class struggle, wage-profit redistribution
Juglar	~9 years	$\gtrless 0$ (oscillating)	Investment cycles, real capital creation/destruction
Long wave	~50 years	$\searrow 0$ (secular)	Rising κ (Marx) + rising ground rent (Ricardo)

The ecological crisis is the second law asserting itself at the system's natural boundary: the Ricardian source of $\dot{\mathcal{H}} \rightarrow 0$ is the thermodynamic limit imposed by a finite entropy sink.

8 Conclusion

We have established a complete two-law thermodynamic framework for classical political economy. The results are summarised in Table 1.

Concept	Classical economics	Thermodynamics
Conservation law	$\Pi = S$ (MELT identity, any price system)	First law: energy conserved
Irreversibility	Abstract labour (metabolic expenditure)	Second law: entropy production
Conversion factor	MELT $\mu = Y/L$	Boltzmann constant k_B
Conservative dynamics	Goodwin, FS, Foley ($\dot{H} = 0$, simple repro)	Closed Hamiltonian system
Non-conservative dynamics	Expanded repro ($\dot{\mathcal{H}} = s_c \varepsilon L / (1 + \varepsilon)$)	Open system with external drive
Economic power	$\dot{\mathcal{H}} = s_c S_\ell$ (surplus labour reinvested)	Power = energy per unit time
Secular limit	$\dot{\mathcal{H}} \rightarrow 0$ as $r^* \rightarrow 0$	Maximum entropy state

The three key results:

1. The unified $\dot{\mathcal{H}}$ formula. All three classical models — Goodwin (class dynamics), Flaschel-Semmler (sectoral dynamics), Foley (circuit of capital) — yield $\dot{\mathcal{H}} = s_c \varepsilon L / (1 + \varepsilon)$ in expanded reproduction. This is not a coincidence but a theorem: all three describe the same underlying circuit, and the MELT is the invariant that bridges them. The exploitation rate ε is the thermodynamic intensity of accumulation; s_c is the fraction of the thermodynamic drive that is converted into capital expansion.

2. The Hamiltonian interpretations. In Goodwin: H is the intensity of class struggle (wage-momentum energy + employment-position energy); $-\partial H / \partial q = -c + d\lambda$ is the Phillips curve force driving wage-share momentum. In FS: H is the intensity of inter-sectoral competitive tension (kinetic price-momentum energy + potential quantity-position energy). In Foley: $\mathcal{H} = K/\mu$ is total capital in labour-time space — economic energy in its thermodynamic substrate.

3. The secular entropy maximum. The long-wave secular decline in r^* has two thermodynamic sources: the Marxian (rising κ displaces living labour) and the Ricardian (exhaustion of natural entropy sinks raises ground rent). Both drive $\dot{\mathcal{H}} \rightarrow 0$. The ecological crisis is the second law asserting itself at the system's natural boundary.

Mirowski's demand for a genuine economic conservation law is met — but only by thermodynamics, not mechanics. The neoclassical programme attempted a mechanics analogy for a thermodynamic

system: it found a conservative field (Hicksian expenditure function) but only in unobservable, distributionally-frozen space. Classical political economy, grounded in the irreversible expenditure of living labour, delivers both laws.

References

Shaikh, Anwar. 2016. *Capitalism: Competition, Conflict, Crises*. Oxford University Press.